

Rabies

Considerations

- Almost always fatal diagnosis frequently made postmortem
- The more the innervation of the area of the bite, the higher the risk
- The closer to the brain the bite, the faster the presentation of rabies
- Average incubation period: a few days but can be several years

Initiation of prophylaxis

- Bite or salivary exposure from bat, or mammalian carnivore
- Close proximity with unknown salivary exposure e.g. live dead bat in room
- Prophylaxis not needed if non salivary exposure, or if bird, reptile, or rodent rare cases

Clinical Rabies

Encephalitis

- Hyper-excitability, disorientation, bizarre behavior, hallucinations
- Autonomic dysfunction e.g. hypersalivation

Paralytic Form

- Paralysis in bitten extremity, quadriplegia, bilateral facial weakness, coma, organ failure, hydrophobia, spasms of larynx, pharynx, diaphragm
- Aerophobia spasms of larynx, pharynx, diaphragm in response to drafts of air = pathognomonic
- Death from various complications: pituitary dysfunction, respiratory dysfunction, cardiac dysfunction, autonomic dysfunction.

EMT/Paramedic

- Follow General Medical Care Guideline
- Follow Bites-Stings Guideline
- Clean the wound thoroughly (copious irrigation)
- Determine Rabies Vaccination status (<3 months prior or > 3 months prior) and Tetanus Vaccination status (<10 years)
- Determine if immunocompetent or immunocompromised
- Determine timeline and risk of exposure
 - Requires assessment for prophylaxis if:
 - Woke up with bat in sleeping quarters
 - Bite from bat or mammalian carnivore (non-domesticated and unvaccinated)
- If clinical rabies is present - take precautions to avoid injury or contact with saliva
 - Follow Behavioral emergency guidelines
 - Follow Seizure guidelines
- Evacuate if
 - Open wound - especially if face, hands, feet, genitalia, or near joint
 - Requiring rabies prophylaxis (see guidelines below)
 - If patient has not had a tetanus vaccine in past 10 years
 - If showing signs of clinical rabies

Algorithm for Prophylaxis

Animal was captured

Yes: Observe 10 days, if abnormal, euthanize and treat patient with the vaccine and RIG. Then patient treatment can be discontinued if the animal pathology is negative for rabies

No: Euthanize animal and treat patient with vaccine and RIG. Discontinue if animal pathology is negative for rabies

Animal was not captured

Yes: Give vaccine and RIG only if rabies risk for species in area

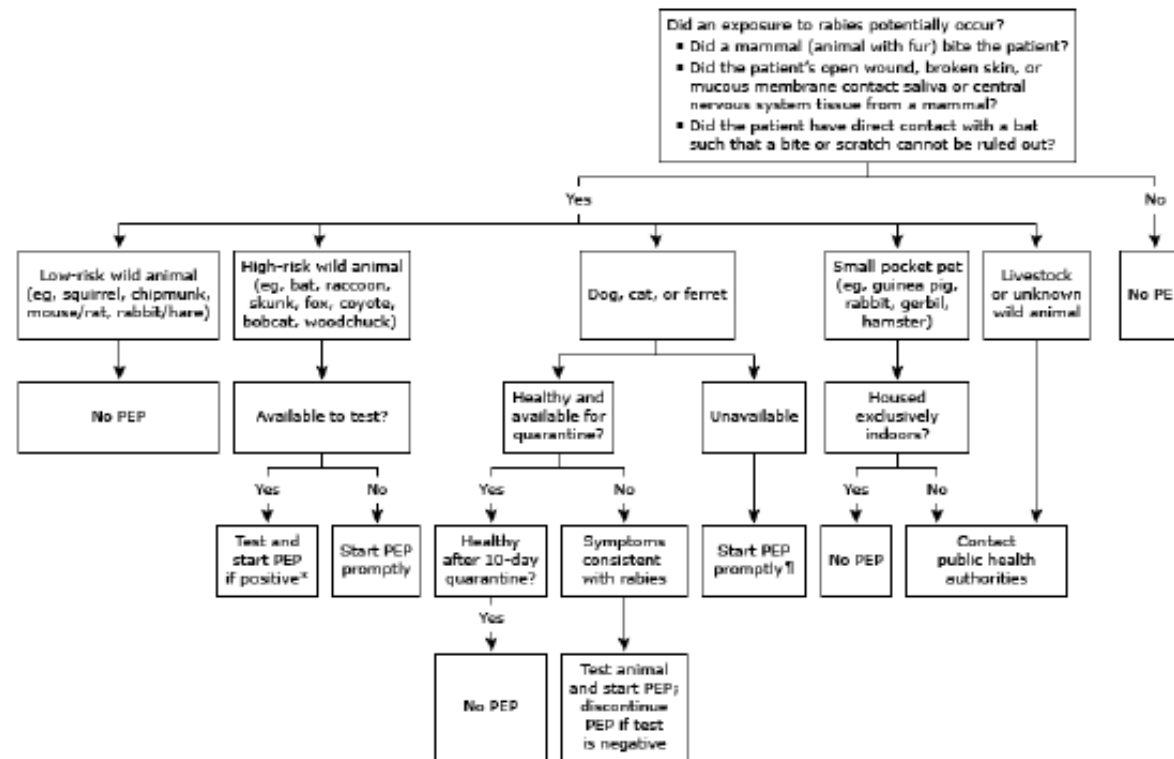
No: Treat with full course vaccine and RIG

Prophylaxis Guidelines (see algorithms below)

Prophylaxis will not be available in the field but guides evacuation measures and timelines

- **Immunocompetent + Vaccinated < 3 months prior - No post-exposure vaccination required**
- Immunocompetent + Vaccinated > 3 months prior - IMMEDIATE evacuation for day 0 vaccination
- Immunocompetent + NOT Vaccinated IMMEDIATE evacuation for day 0 vaccination
- Immunocompromised + either Vaccinated **or** NOT Vaccinated - IMMEDIATE evacuation

Rabies postexposure prophylaxis algorithm



PEP: postexposure prophylaxis.

* PEP should be initiated immediately in patients with severe bites to the head, neck, or trunk after an unprovoked attack from a high-risk animal. PEP can be discontinued if testing proves the animal was not rabid.

† In areas without an endemic, terrestrial strain of rabies (eg, dog, raccoon, skunk, fox, mongoose), contact local public health authorities for risk assessment.

Courtesy of Alfred DeMaria Jr, MD and the Massachusetts Department of Public Health.

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World Health Organization post-exposure rabies management

Category	PEP vaccine regimen	RIG	Comments
Category I – Touching or feeding animals, animal licks on intact skin	Not required	Not required	Exposed skin surfaces should be washed
Category II – Nibbling of uncovered skin, minor scratches or abrasions without bleeding	Patients who have NOT been previously immunized: Immediate vaccination with one of the following regimens: 2-sites ID on days 0, 3, and 7 1-site IM on days 0, 3, 7, and between days 14 and 28 2-sites IM on day 0 and 1-site IM on days 7 and 21	Not required	Exposed skin surfaces should be washed
	Patients who have been previously immunized:* 1-site ID on days 0 and 3 4-sites ID on day 0 1-site IM on days 0 and 3	Not required	Exposed skin surfaces should be washed
Category III – Single or multiple transdermal bites or scratches, contamination of mucous membrane or broken skin with saliva from animal licks, exposures due to direct contact with bats	Patients who have NOT been previously immunized: Immediate vaccination with one of the following regimens: 2-sites ID on days 0, 3, and 7 1-site IM on days 0, 3, 7, and between days 14 and 28 2-sites IM on day 0 and 1-site IM on days 7 and 21	RIG should be administered [¶]	Exposed skin surfaces should be washed
	Patients who have been previously immunized:* 1-site ID on days 0 and 3 4-sites ID on day 0 1-site IM on days 0 and 3	Not required	Exposed skin surfaces should be washed

This table describes regimens recommended by the World Health Organization (WHO) for post-exposure prophylaxis after a potential rabies exposure and are intended for areas that may have limited resources. Refer to the topic that discusses rabies vaccine and immunoglobulin for information on regimens recommended by the Centers for Disease Control and Prevention's Advisory Committee on Immunization Practices (ACIP) for use in the United States, as well as information on vaccine and immunoglobulin formulations and administration.

Advisory Committee on Immunization Practices: ACIP; ID: intradermal; IM: intramuscular; PEP: post-exposure prophylaxis; RIG: rabies immune globulin.

* Previously immunized patients include those who have received pre-exposure prophylaxis and those who have completed a post-exposure vaccine regimen >3 months ago. Post-exposure prophylaxis (except for wound care) is not recommended for individuals who have a rabies exposure <3 months after completing a post-exposure vaccine series.

¶ RIG should be infiltrated into and around the wound. For small wounds, the maximal quantity that is anatomically feasible should be administered. Similarly, for bat bites or scratches, which are not readily visible, RIG should be injected around the site of the exposure to the degree that is anatomically feasible. The WHO no longer recommends injecting the remainder of the RIG IM at a site distant from the wound, which differs from ACIP recommendations in the United States. Refer to the topic that discusses rabies immunoglobulin for ACIP recommendations.

Adapted from: Ridder BA. Rabies vaccines: WHO position paper – April 2018. *Weekly Epidemiological Record* 2018; 93(16):201-220. Available at: <https://www.who.int/publications/i/item/who-wer9316> (Accessed on December 10, 2021).

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